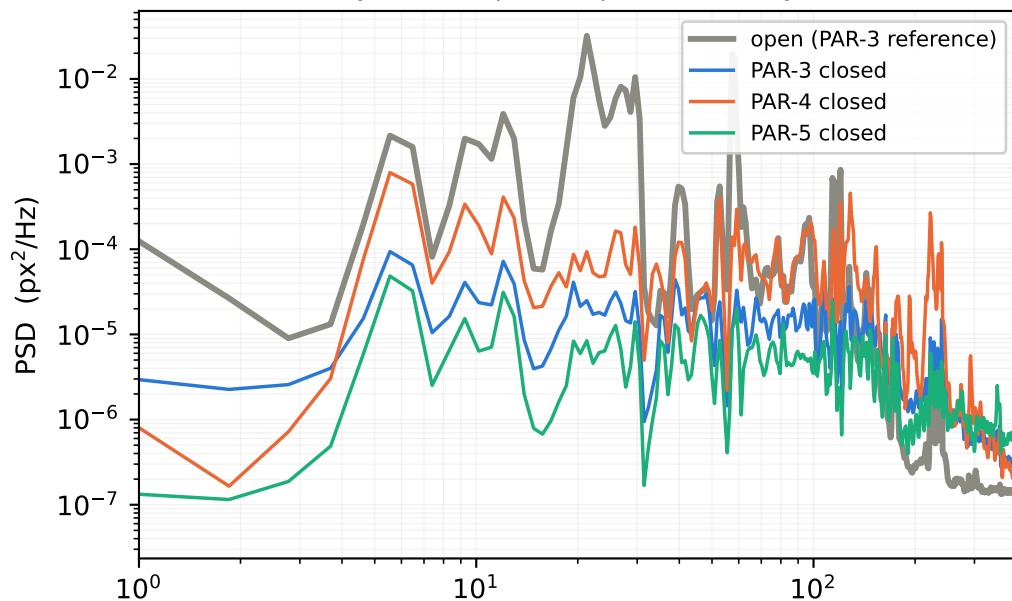
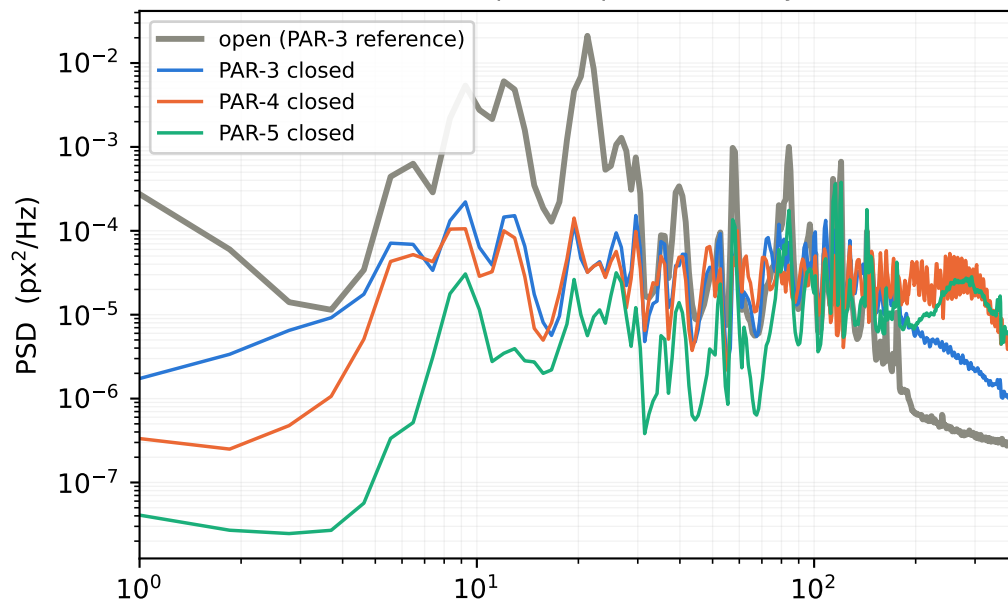


PAR-5 closed loop vs PAR-3 reference open loop
PAR-5's own open phase is discarded (mount disturbance pinned the beam at the ROI edge, $t \approx 200\text{-}600\text{ s}$)

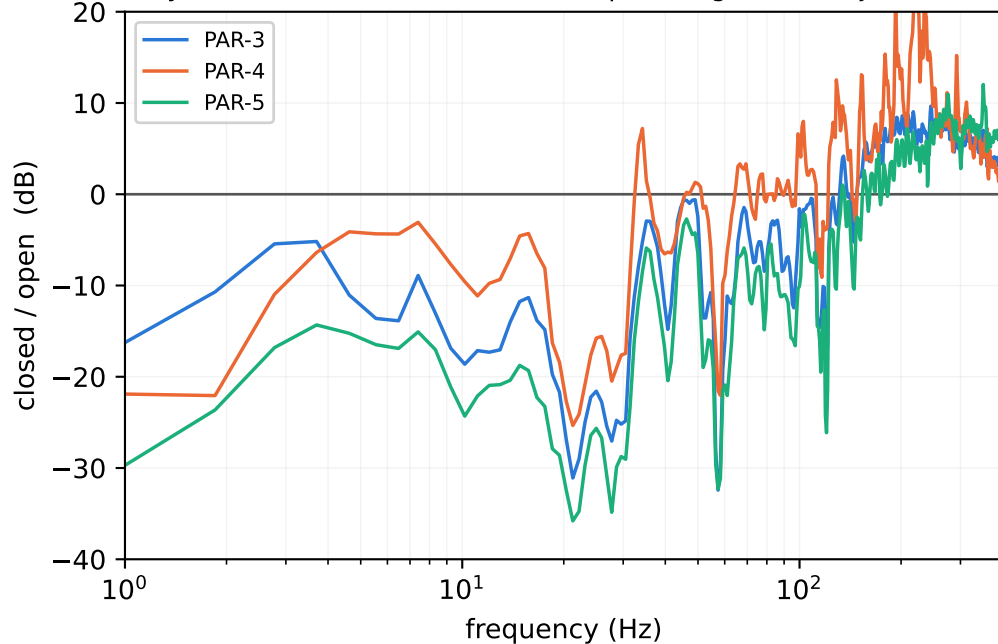
y axis — power spectral density



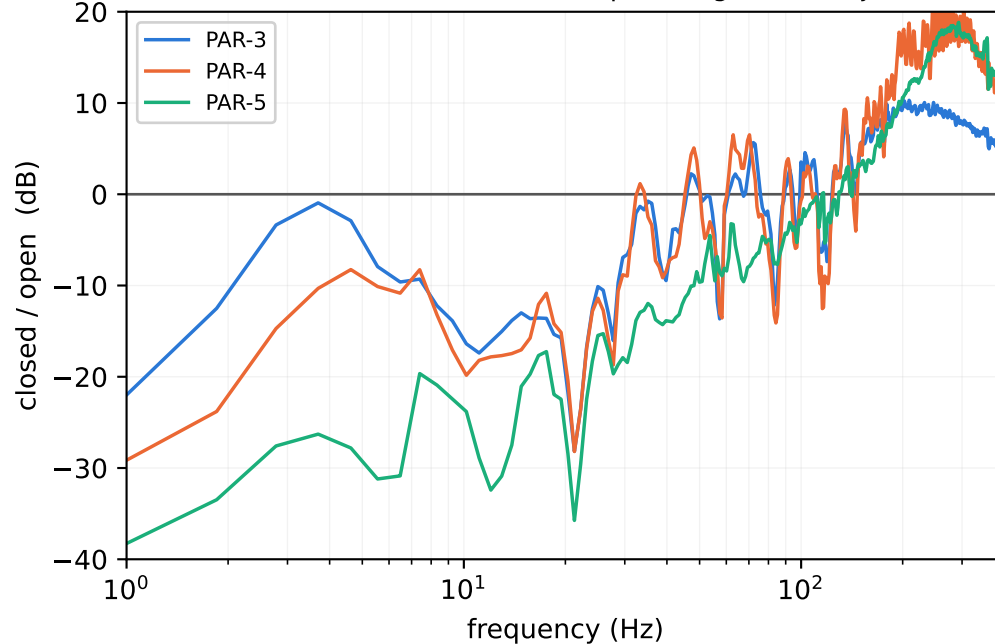
x axis — power spectral density



y axis — attenuation vs PAR-3 open (negative = rejection)



x axis — attenuation vs PAR-3 open (negative = rejection)



PAR-5 SPLICED BAND BUDGET (RMS in px; open = PAR-3 reference)

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--- y axis ---

band (Hz)	open	PAR-3	PAR-4	PAR-5	PAR-5 rej	
1-10	0.0706	0.0146	0.0403	0.0098	7.2x	
10-30	0.3265	0.0206	0.0435	0.0117	27.8x	
30-60	0.2042	0.0208	0.0441	0.0136	15.0x	
60-150	0.0886	0.0361	0.0854	0.0229	3.9x	
150-250	0.0110	0.0204	0.0534	0.0144	0.8x	
250-400	0.0049	0.0095	0.0113	0.0116	0.4x	
1-400	0.4137	0.0544	0.1278	0.0364	11.4x	<- TOTAL

--- x axis ---

band (Hz)	open	PAR-3	PAR-4	PAR-5	PAR-5 rej	
1-10	0.0766	0.0204	0.0167	0.0059	13.1x	
10-30	0.2478	0.0329	0.0290	0.0136	18.2x	
30-60	0.0599	0.0296	0.0278	0.0191	3.1x	
60-150	0.0839	0.0605	0.0507	0.0585	1.4x	
150-250	0.0126	0.0305	0.0499	0.0333	0.4x	
250-400	0.0071	0.0169	0.0499	0.0475	0.2x	
1-400	0.2871	0.0869	0.0984	0.0863	3.3x	<- TOTAL

SENSITIVITY TO THE OPEN REFERENCE (total 1-400 Hz)

axis	open ref	open px	PAR-5 closed	rejection
y	PAR-3	0.4137	0.0364	11.4x
x	PAR-3	0.2871	0.0863	3.3x
y	PAR-4	0.4925	0.0364	13.5x
x	PAR-4	0.3962	0.0863	4.6x

TIME-DOMAIN RMS from each run's own '# stat' line (open mean / open rms / closed rms, px):

PAR-3:	y park	-1.74	open 0.4685	closed 0.0565		x park	+3.53	open 0.3162	closed 0.0891
PAR-4:	y park	+5.95	open 0.5525	closed 0.1285		x park	+0.44	open 0.4225	closed 0.1020
PAR-5:	y park	+3.49	open 2.5128	closed 0.0394		x park	+10.91	open 6.6998	closed 0.0914

CAVEATS

- * OPEN REFERENCE IS PAR-3, chosen for ENVIRONMENT match rather than recency: PAR-5's own clean pre-bump window (t=110-200 s) read y 0.456 / x 0.339 px, against PAR-3's y 0.465 / x 0.312 and PAR-4's y 0.553 / x 0.423. PAR-4 is the same day but its open is 21% (y) / 25% (x) larger, which would flatter PAR-5's ratio. The PAR-4-referenced numbers are in the sensitivity block above -- quote the pair, the choice moves the headline 15-40%.
- * PAR-3 scored t=720-1220 s (settle_time 120) while still converging; PAR-4 and PAR-5 scored t=1200-1700 (settle_time 600). PAR-3's closed trace is pessimistic, not a fair peer.
- * PAR-4 ran with the beam parked +5.95 px, which pushed the 20x20 CoM window off the 32x32 ROI edge and desensitized the y centroid ~2x. Its y trace is not trustworthy.
- * PAR-5's own open PSD is discarded here; its '# stat' open figures (y 2.51 / x 6.70 px) include the mount disturbance and must not be quoted.
- * PAR-5's raw error/command records, weights and convergence trace were overwritten at 14:12 by a re-run using the same output filenames. Only the .dat/.pdf survive.